



AAI presents IUT 12th ICT Fest
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Bdapps presents
Agentic AI Hackathon
powered by Codex

Mock Round Problem Statement

Bug Fix Challenge — Todo REST API

Repository Link

[Agentic AI Hackathon Mock Repository](#)

Overview

This is a **bug fix challenge**. Participants are given a broken codebase and must find the bugs, understand why they are broken, and fix them. There are **4 bugs** hidden across the project, ranging from easy one-liners to slightly more subtle logic issues.

Participants do *not* need to add features or refactor anything. Find the bugs. Fix them. That is the entire task.

The Project

A simple **Todo REST API** built with [FastAPI](#).

Users can register, log in, and manage their own todo items. Each todo has a title, description, deadline, and priority level. Authentication is handled via JWT tokens.

File Structure

```
app/
|-- main.py          # API route handlers
|-- auth.py          # JWT creation, password hashing, auth dependency
|-- models.py        # SQLAlchemy database models
|-- schemas.py       # Pydantic request/response schemas
'-- database.py      # Database engine and session setup
```

API Endpoints

Method	Path	Auth	Description
POST	/auth/register	No	Register a new user
POST	/auth/login	No	Log in, receive a JWT token
POST	/todos	Yes	Create a todo
GET	/todos	Yes	List your todos
GET	/todos/{id}	Yes	Get a single todo
PUT	/todos/{id}	Yes	Update a todo
DELETE	/todos/{id}	Yes	Delete a todo
GET	/health	No	Health check

For all `/todos` endpoints, pass the token in the `Authorization` header:

Authorization: Bearer <your_token>

Running the Project

With Docker (recommended)

```
docker compose up --build
```

Without Docker

```
python -m venv .venv
```

```
# Windows
```

```
.venv\Scripts\activate
```

```
# macOS / Linux
```

```
source .venv/bin/activate
```

```
pip install -r requirements.txt
```

```
uvicorn app.main:app --reload
```

App runs at <http://localhost:8000>. Interactive API docs at <http://localhost:8000/docs>.

How to Test

Use the Swagger UI at `/docs`, `curl`, or Postman.

Postman Setup

1. Register: `POST /auth/register` → Body: *raw / JSON* →

```
{"username": "alice", "password": "pass123"}
```
2. Login: `POST /auth/login` → Body: *x-www-form-urlencoded* → keys `username` and `password`
3. Copy the `access_token` from the login response
4. For all todo requests: Authorization tab → Bearer Token → paste token

curl Example

```
# Register
```

```
curl -X POST http://localhost:8000/auth/register \
  -H "Content-Type: application/json" \
  -d '{"username": "alice", "password": "pass123"}'
```

```
# Login
```

```
curl -X POST http://localhost:8000/auth/login \
  -d "username=alice&password=pass123"
```

```
# Use the token
```

```
curl http://localhost:8000/todos \
  -H "Authorization: Bearer <TOKEN>"
```

The Challenge

There are **4 bugs** in the codebase. Each bug causes clearly observable, wrong behavior when interacting with the API.

What counts as a valid fix:

- The fixed code produces the correct behavior described for that bug
- Only the broken code should be changed — do not refactor or rewrite unrelated code

Submission

This is the mock round — no submission required. Use this round to familiarize yourself with the format. Official answers are provided in `bugs.md`.

The repository for this mock round is available below — fork it, explore the codebase, and try finding and fixing all 4 bugs on your own before checking `bugs.md`.

Mock round repository:
[Agentic AI Hackathon Mock Repository](#)

For Reference: Preliminary Round Submission

The actual **preliminary round** will follow the same format as this mock round, with a new repository and a new set of bugs. Submission for the preliminary round will work as follows:

1. Fork the preliminary-round repository to your own GitHub account
2. Fix the bugs in your fork
3. Submit your repository URL via the Google Form link that will be provided on the day of the competition

Repository Visibility Rules

- Your repository may be kept **private** during the competition
- You must make it **public within 1 hour** of the competition ending — repositories that remain private after this window will not be evaluated

Scoring & Tie-Breaking

Shown here for reference — this applies to the preliminary round, not the mock round, since no submission is required for the mock round.

Points are awarded per bug based on difficulty:

Difficulty	Points
Easy	3
Medium	5
Hard	10

Tie-Breaking

Ties are resolved in the following order:

1. **Difficulty of bugs solved** — the participant who fixed harder bugs ranks higher
2. `bug_report.md` (**optional**) — if a tie remains after step 1, participants who submitted a `bug_report.md` in the root of their repository go through manual evaluation

`bug_report.md` should include, for each bug found:

- Which file(s)/line(s) the bug is on
- What the bug was and why it caused incorrect behavior
- How it was fixed

Manual evaluation of `bug_report.md` is the final tie-breaking mechanism.

Good luck.